

**Test Details**Total Number of Question: **45**Total Duration: **60 mins****Click on the button below to start the test**

Select the section you want to attempt

Section Name	Number of Question
☒ AR	15
☐ OOPS	14
☐ Data Structure	16

Start test

WARNING

All Question of this Section are Required to be Attempted in Serial Number.

OK

WARNING

All Question of this Section are Required to be Attempted in Serial Number.

OK

WARNING

All Question of this Section are Required to be Attempted in Serial Number.

OK

Time Left: 56:07

Section 1 of 3

AR



Answered	1
Revisit	0
Not Answered	14

Finish Test

Question # 2

Revisit

The problem below consists of a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statements is sufficient to answer the question. Read both the statements and give the answer:

Who among L, N, F, G and Q was the first to reach the college?

I. F reached before L and G but not before Q who was not the first to reach.

II. N reached before F and G but L reached after F.

- ☒ A - Statements i and ii together also are not sufficient.
- ☐ B - Statement ii alone is sufficient
- ☐ C - Statements i and ii together are sufficient.
- ☐ D - Statement i alone is sufficient
- ☐ E - Either of the statements i or ii alone is sufficient.

Clear Response



2/15



WARNING

All Question of this Section are Required to be Attempted in Serial Number.

OK

Time Left: 49:30

Section 1 of 3

AR



Answered	6
Revisit	0
Not Answered	9

Finish Test

Question # 7

Revisit

Fill in the blank with the best suited word.

In this biography we are given a glimpse of the young man ____ (i)____ pursuing the path of the poet despite ____ (ii)____ and rejection slips.

- ☐ A - (i) Indefatigably ; (ii) Disappointment
- ☐ B - (i) Insignificantly ; (ii) Plaudits
- ☐ C - (i) Desultorily ; (ii) Exhortation



7/15



Time Left: 47:24

Section 1 of 3

AR



Answered	6
Revisit	0
Not Answered	9

Finish Test

Question # 7

Revisit

Fill in the blank with the best suited word.

In this biography we are given a glimpse of the young man ____ (i)____ pursuing the path of the poet despite ____ (ii)____ and rejection slips.

- ☐ A - (i) Indefatigably ; (ii) Disappointment
- ☐ B - (i) Insignificantly ; (ii) Plaudits
- ☐ C - (i) Desultorily ; (ii) Exhortation



7/15



Time Left: 18:34

Section 3 of 3

Data Structure [Finish Test](#)

- 1
- 2
- 3
- 4
- 5
- 6
- 7
- 8
- 9
- 10
- 11
- 12
- 13
- 14
- 15
- 16

Answered	1
Revisit	0
Not Answered	15

Question # 2

[Revisit](#)

Consider a directed graph G with distinct and non-negative edge lengths and a source vertex S . Which of the following statements about a fixed destination vertex T , assuming that the graph contains at least one $S \sim T$ path, is/are true?

- ☐ A - The shortest $S \sim T$ path must exclude the maximum-length edge of G .
- ☐ B - The shortest $S \sim T$ path must include the minimum-length edge of G .
- ☐ C - All of the given options
- ☐ D - The shortest $S \sim T$ path might have as many as $n-1$ edges, where n is the number of vertices.



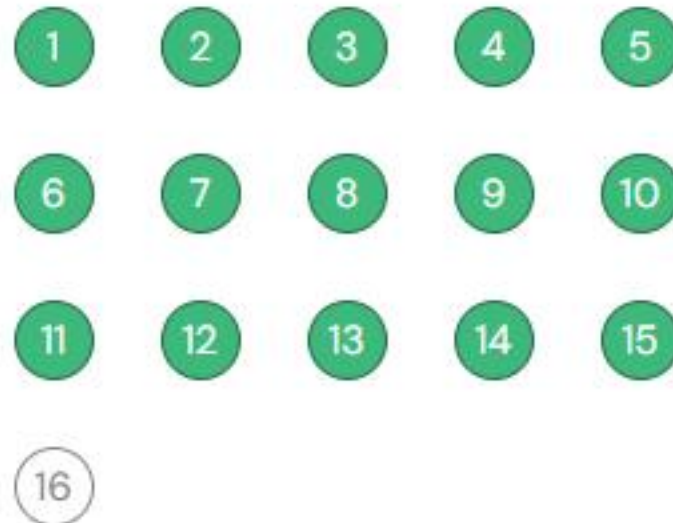
2/16



Time Left: 10:27

Section 3 of 3

Data Structure



Answered	15
Revisit	0
Not Answered	1

Question # 15

Finish Test

Revisit

Choose the right statements that hold good for the given code.

```
#include <stdio.h>
int* fun_arr(){
    static int arr1[20];
    arr1[0]=1;
    arr1[1]=2;
    return arr1;
}
int main(){
    int* ptr1=fun_arr();
    printf("%d %d", ptr1[0], ptr1[1]);
    return 0;
}
```

- ☐ A - 1,4
- ☒ B - 1,2
- ☐ C - 1,2,3,4
- ☐ D - 3,4



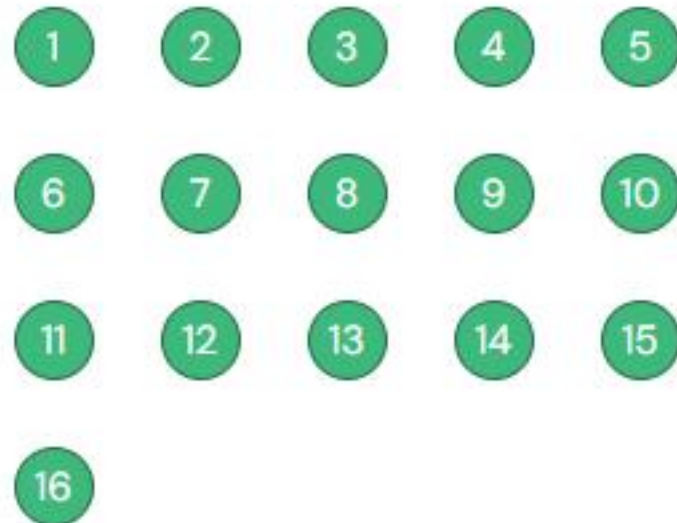
15/16



Time Left: 08:04

Section 3 of 3

Data Structure ▾



Answered	16
Revisit	0
Not Answered	0

Finish Test

Question # 2

 Revisit

Consider a directed graph G with distinct and non-negative edge lengths and a source vertex S . Which of the following statements about a fixed destination vertex T , assuming that the graph contains at least one $S \sim T$ path, is/are true?

- ☐ A - The shortest $S \sim T$ path must exclude the maximum-length edge of G .
- ☒ B - The shortest $S \sim T$ path must include the minimum-length edge of G .
- ☐ C - All of the given options
- ☐ D - The shortest $S \sim T$ path might have as many as $n-1$ edges, where n is the number of vertices.

Clear Response



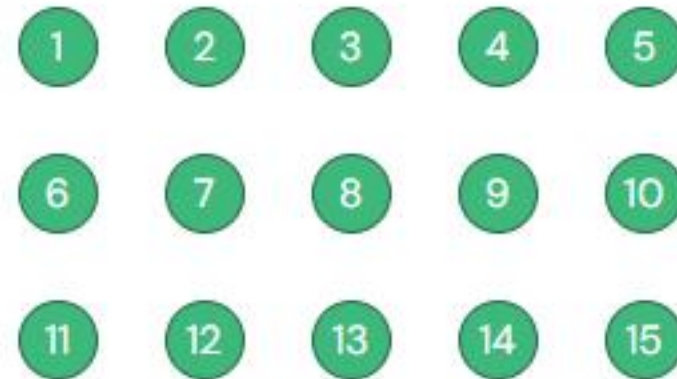
2/16



Time Left: 06:04

Section 1 of 3

AR



Answered	15
Revisit	0
Not Answered	0

Finish Test

Question # 13

Revisit

Choose the option for which the analogy between the pair of words is the same as the one given in the question.
VERNACULAR : REGIONAL ::

- ☐ A - fluctuation : irregular
- ☐ B - budget : austere
- ☒ C - leaf : autumnal
- ☐ D - senator : dissentious

Clear Response



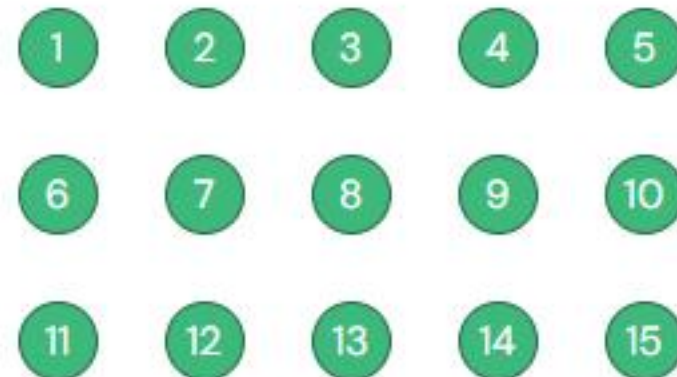
13/15



Time Left: 05:45

Section 1 of 3

AR



Answered	15
Revisit	0
Not Answered	0

Finish Test

Question # 14

Revisit

Choose the option for which the analogy between the pair of words is the same as the one given in the question.
BISHOP : ORDAINED ::

- ☒ A - prince : opulent
- ☐ B - cliché : intrepid
- ☐ C - loft : unattainable
- ☐ D - transgression : acceptable

Clear Response



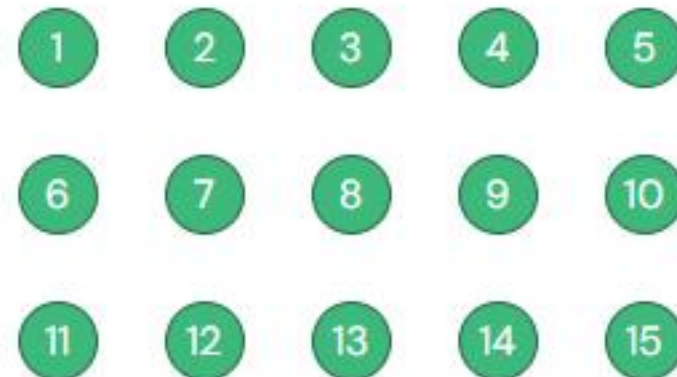
14/15



Time Left: 06:17

Section 1 of 3

AR



Answered	15
Revisit	0
Not Answered	0

Finish Test

Question # 10

Revisit

Choose the option for which the analogy between the pair of words is the same as the one given in the question.
NOISOME : GARBAGE ::

- ☐ A - liquid : perfume
- ☐ B - heavy : metal
- ☒ C - warmth : snow
- ☐ D - fragrant : incense

Clear Response



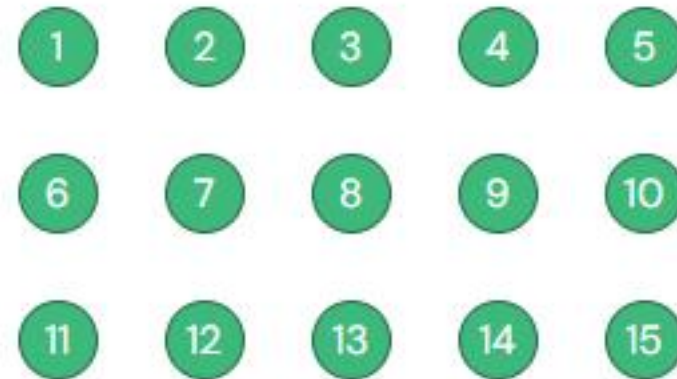
10/15



Time Left: 04:53

Section 1 of 3

AR



Answered	15
Revisit	0
Not Answered	0

Finish Test

Question # 15

Revisit

Choose the option for which the analogy between the pair of words is the same as the one given in the question.
APIARY : BEE ::

- ☐ A - arboretum : tree
- ☒ B - dam : water
- ☐ C - forum : speech
- ☐ D - museum : painting

Clear Response



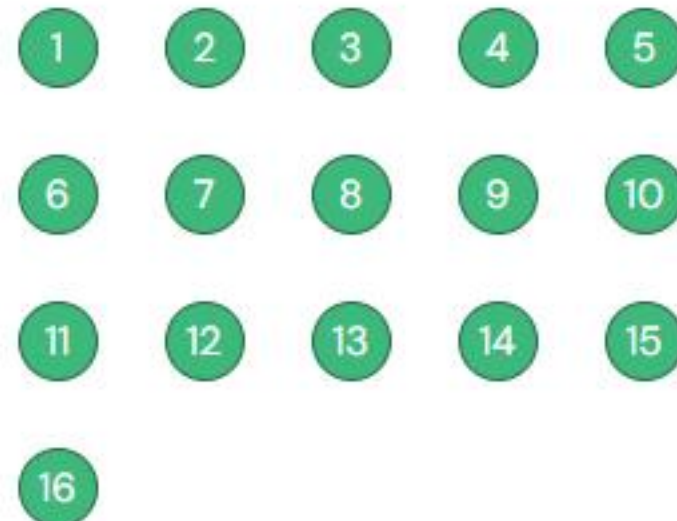
15/15



Time Left: 03:05

Section 1 of 3

Data Structure



Answered	16
Revisit	0
Not Answered	0

Finish Test

Question # 2

Revisit

Consider a directed graph G with distinct and non-negative edge lengths and a source vertex S . Which of the following statements about a fixed destination vertex T , assuming that the graph contains at least one $S \sim T$ path, is/are true?

- ☐ A - The shortest $S \sim T$ path must exclude the maximum-length edge of G .
- ☒ B - The shortest $S \sim T$ path must include the minimum-length edge of G .
- ☐ C - All of the given options
- ☐ D - The shortest $S \sim T$ path might have as many as $n-1$ edges, where n is the number of vertices.

Clear Response



2/16





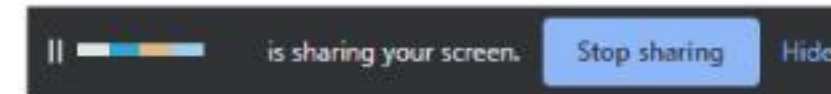
Test Details

Total Number of Question: **45**

Total Duration: **60 mins**

3. Your test opens in a new popup window once you press the launch test button.

4. Before proceeding for the next step, please hide the **Screen Share** overlay (shown below) by clicking on the **"Hide"** button so that the proctoring won't detect it as **"Navigated Away"** event.



Section Wise Details

Section Name	Number of Question
AR	15
OOPS	14
Data Structure	16

**Test Details**Total Number of Question: **45**Total Duration: **60 mins****Click on the button below to start the test**

Select the section you want to attempt

Section Name	Number of Question
☒ AR	15
☐ OOPS	14
☐ Data Structure	16

Start test

Section 1 of 3

AR



Answered	0
Revisit	1
Not Answered	15

Question # 2

Revisit

The problem below consists of a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statements is sufficient to answer the question. Read both the statements and give the answer:

Who among L, N, F, G and Q was the first to reach the college?

I. F reached before L and G but not before Q who was not the first to reach.

II. N reached before F and G but L reached after F.

- ☐ A – Statements i and ii together also are not sufficient.
- ☐ B – Statement ii alone is sufficient
- ☐ C – Statements i and ii together are sufficient.
- ☐ D – Statement i alone is sufficient
- ☐ E – Either of the statements i or ii alone is sufficient.



WARNING

All Question of this Section are Required to be Attempted in Serial Number.

OK

Time Left: 59:46

Section 1 of 3

AR



- 1 2 3 4 5
6 7 8 9 10
11 12 13 14 15

Answered	0
Revisit	0
Not Answered	15

Question # 1

Revisit

In a certain code the symbol for 0 (zero) is I and that for 1 is H. There are no other symbols for numbers and all numbers greater than 1 are written using these two symbols only, the value of the symbol for 1 doubling itself every-time it shifts one place-value (ones,tens, hundreds and so on) to the left.

Thus—

0 is written I

1 is written H

2 is written H I

3 is written H H

4 is written H I I and so on

Which of the following numbers is written as H I H I in that code?

- ☐ A – 8080
☐ B – 202
☐ C – 42
☐ D – 10
☐ E – 9



1/15



Time Left: 58:25

Section 1 of 3

AR



Answered	0
Revisit	3
Not Answered	15

Question # 4

Revisit

The problem below consists of a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statements is sufficient to answer the question. Read both the statements and give the answer:

In the code language what is the code for 'fat'?

I. In the code language 'she is fat' is written as 'he ra ca'.

II. In the same code language 'fat boy' is written as 'ra ka'.

- ☐ A – Statements i and ii together also are not sufficient.
- ☐ B – Statement ii alone is sufficient
- ☐ C – Statement i alone is sufficient
- ☐ D – Either of the statements i or ii alone is sufficient.
- ☐ E – Statements i and ii together are sufficient.



4/15



Time Left: 58:12

Section 1 of 3

AR



Finish Test

Question # 5

Revisit

Fill in the blank with the best suited word.

A businessman must _____(i)_____ his horizons; a/an _____(ii)_____ attitude will get you nowhere in this age of global communications.

☒ A – (i) Broaden ; (ii) Bigoted☐ B – (i) Restrain ; (ii) Extensive☐ C – (i) Envisage ; (ii) Middling

Clear Response

Answered

0

Revisit

4

Not Answered

15



5/15



Time Left: 58:40

Section 1 of 3

AR



1	2	3	4	5
6	7	8	9	10
11	12	13	14	15

Answered	0
Revisit	2
Not Answered	15

Question # 3

Revisit

The problem below consists of a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statements is sufficient to answer the question. Read both the statements and give the answer:

What is the value of $36\$4H8$?

I. $P\$Q$ means divide P by Q.

II. AHB means multiply A by B.

- ☐ A – Statements i and ii together also are not sufficient.
- ☐ B – Either of the statements i or ii alone is sufficient.
- ☐ C – Statement i alone is sufficient
- ☐ D – Statement ii alone is sufficient
- ☐ E – Statements i and ii together are sufficient.



3/15



Time Left: 57:48

Section 1 of 3

AR



Finish Test



Answered	1
Revisit	5
Not Answered	14

Question # 7

Revisit

Fill in the blank with the best suited word.

The game of chess is an example of a information system: the pieces sit inertly on the board until the players move them according to known rules.

- ☐ A – Synergistic
- ☐ B – Innovative
- ☐ C – Logistic
- ☐ D – Passive



7/15



Time Left: 56:13

Section 1 of 3

AR



Finish Test



Answered	3
Revisit	7
Not Answered	12

Question # 11

Revisit

Radha from point A walked 30Km toward north. Then moved 60km in south from here to reach point B. From here, she turned to her left and walked 10km to reach point C. From the point C, she walked 70km north and turned right to walk another 80km. Finally, she turned to her right and walked 70km before stopping to reach point D. What is the minimum distance between A and D?

- ☐ A – 30Km
- ☐ B – $30\sqrt{10}$ km
- ☐ C – $30\sqrt{15}$ km
- ☐ D – $30\sqrt{5}$ km



11/15



Time Left: 56:31

Section 1 of 3

AR

[Finish Test](#)

Answered	1
Revisit	6
Not Answered	14

Question # 8

[Revisit](#)

If “you can win” is coded as “joe nee sop”, “can you do” is coded as “sop nee tee” and “do you play” is coded as “nee sow tee”, then how is “play can win” coded in that language?

- ☐ A – sop tee joe
- ☐ B – joe tee sow
- ☐ C – nee tee sow
- ☐ D – sop sow joe



8/15



Time Left: 55:43

Section 1 of 3

AR

[Finish Test](#)

Answered	3
Revisit	9
Not Answered	12

Question # 13

[Revisit](#)

Choose the option for which the analogy between the pair of words is the same as the one given in the question.
DELUGE : WATER ::

- ☐ A – sleet : hail
- ☐ B – dew : rain
- ☐ C – igloo : ice
- ☐ D – landslide : earth



13/15



Time Left: 55:34

Section 1 of 3

AR



Finish Test



Answered	3
Revisit	10
Not Answered	12

Question # 14

Revisit

If EXPEDIGNCR is coded as FQYFKFQBMF, then what would be the code of "RESOLUTION" ?

- ☐ A – PTFSWNMNHS
- ☐ B – PTFSWMMNIT
- ☐ C – PUFSWMMNIS
- ☐ D – PTFSXNMNHS



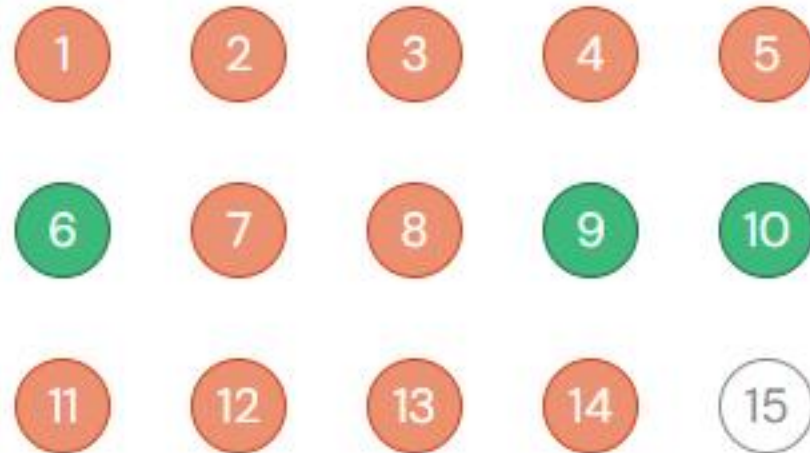
14/15



Time Left: 55:23

Section 1 of 3

AR



Answered	3
Revisit	11
Not Answered	12

Finish Test

Question # 15

Revisit

Choose the alternative that best completes the given comparison.

----- : poetry :: comedy : drama

- ☐ A – ballad
- ☐ B – elegy
- ☐ C – limerick
- ☐ D – sonnet



15/15



Time Left: 55:58

Section 1 of 3

AR



Finish Test



Answered	3
Revisit	8
Not Answered	12

Question # 12

Revisit

If 'A + B' means 'A is the father of B', 'A - B' means 'A is the wife of B', 'A × B' means 'A is the brother of B' and 'A ÷ B' means 'A is the daughter of B' then which of the following relationship is best expressed by the expression 'S is the niece of T' ?

- ☐ A - T × M + S - K
- ☐ B - T × S + M - K
- ☐ C - T + M × S - K
- ☐ D - K - S × M + T



12/15



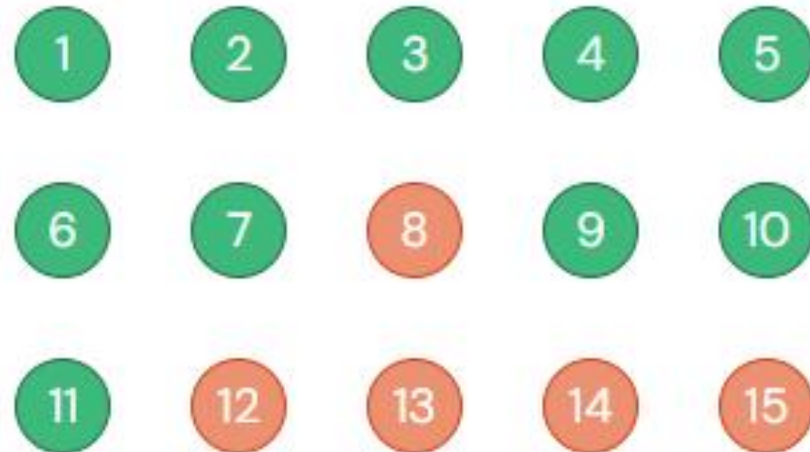
Time Left: 44:09

Section 1 of 3

AR



Finish Test



Answered	10
Revisit	5
Not Answered	5

Question # 14

✕ Revisit

If EXPEDIGNCR is coded as FQYFKFQBMF, then what would be the code of "RESOLUTION" ?

- ☐ A – PTFSWNMNHS
- ☐ B – PTFSWMMNIT
- ☐ C – PUFSWMMNIS
- ☐ D – PTFSXNMNHS



14/15



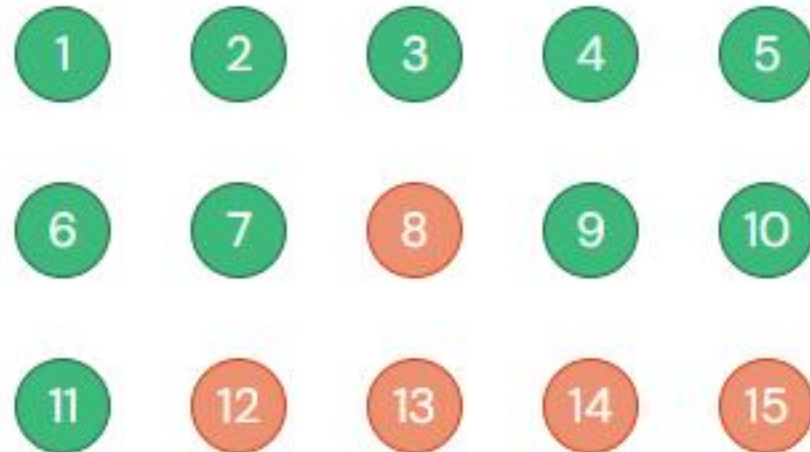
Time Left: 44:19

Section 1 of 3

AR



Finish Test



Answered	10
Revisit	5
Not Answered	5

Question # 13

✕ Revisit

Choose the option for which the analogy between the pair of words is the same as the one given in the question.
DELUGE : WATER ::

- ☐ A – sleet : hail
- ☐ B – dew : rain
- ☒ C – igloo : ice
- ☐ D – landslide : earth

Clear Response



13/15



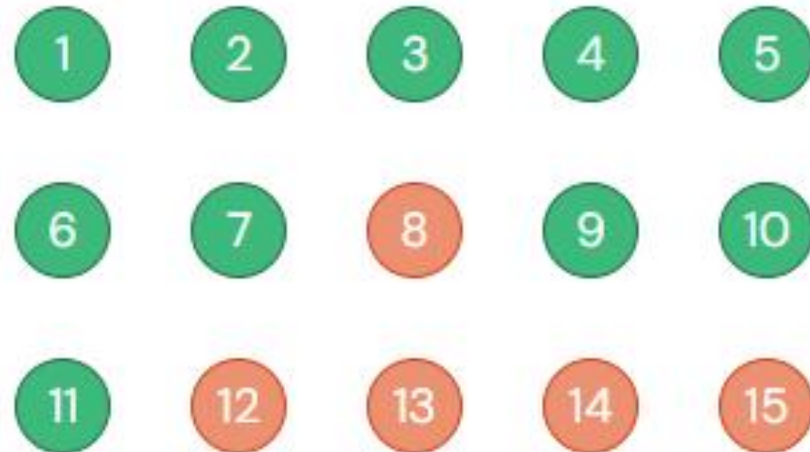
Time Left: 43:53

Section 1 of 3

AR



Finish Test



Answered	10
Revisit	5
Not Answered	5

Question # 12

✕ Revisit

If 'A + B' means 'A is the father of B', 'A - B' means 'A is the wife of B', 'A × B' means 'A is the brother of B' and 'A ÷ B' means 'A is the daughter of B' then which of the following relationship is best expressed by the expression 'S is the niece of T' ?

- ☐ A - T × M + S - K
- ☒ B - T × S + M - K
- ☐ C - T + M × S - K
- ☐ D - K - S × M + T

Clear Response



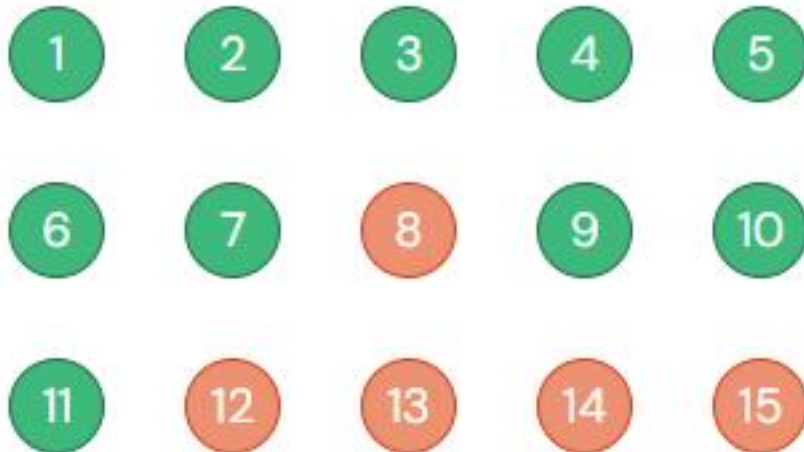
12/15



Time Left: 43:38

Section 1 of 3

AR



Answered	10
Revisit	5
Not Answered	5

Question # 15

[✕ Revisit](#)

Choose the alternative that best completes the given comparison.

_____ : poetry :: comedy : drama

- ☐ A – ballad
- ☐ B – elegy
- ☐ C – limerick
- ☐ D – sonnet



15/15



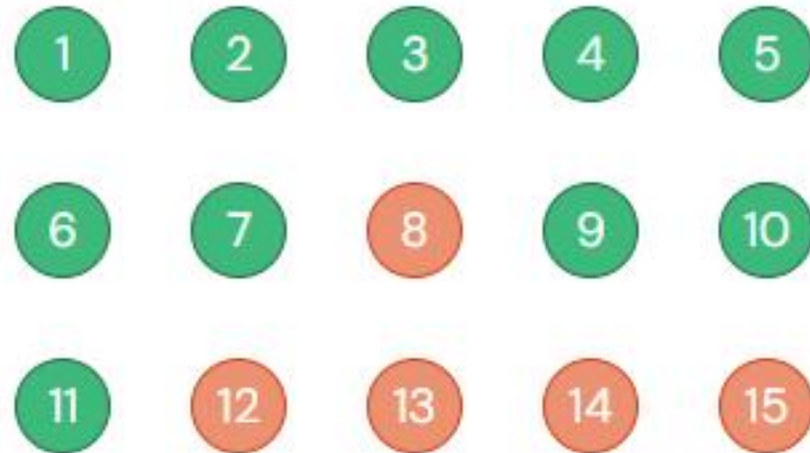
Time Left: 43:34

Section 1 of 3

AR



Finish Test



Answered	10
Revisit	5
Not Answered	5

Question # 14

✕ Revisit

If EXPEDIGNCR is coded as FQYFKFQBMF, then what would be the code of "RESOLUTION" ?

- ☐ A – PTFSWNMNHS
- ☐ B – PTFSWMMNIT
- ☐ C – PUFSWMMNIS
- ☐ D – PTFSXNMNHS



14/15



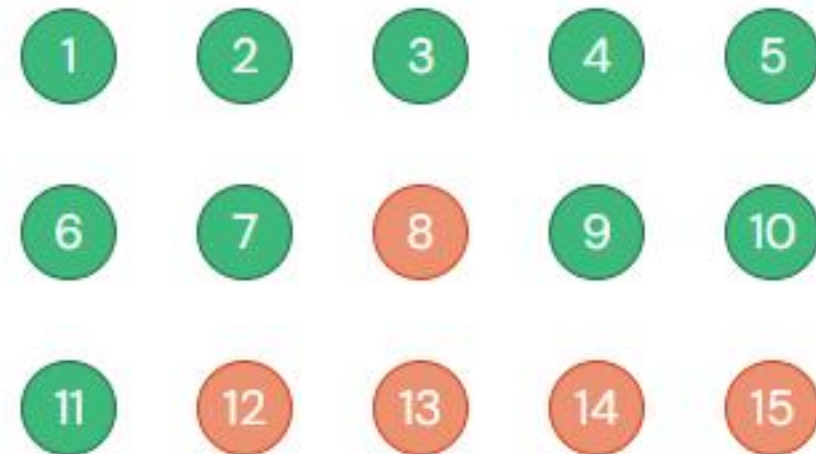
Time Left: 43:29

Section 1 of 3

AR



Finish Test



Answered	10
Revisit	5
Not Answered	5

Question # 8

✕ Revisit

If "you can win" is coded as "joe nee sop", "can you do" is coded as "sop nee tee" and "do you play" is coded as "nee sow tee", then how is "play can win" coded in that language?

- ☐ A – sop tee joe
- ☐ B – joe tee sow
- ☐ C – nee tee sow
- ☐ D – sop sow joe



8/15



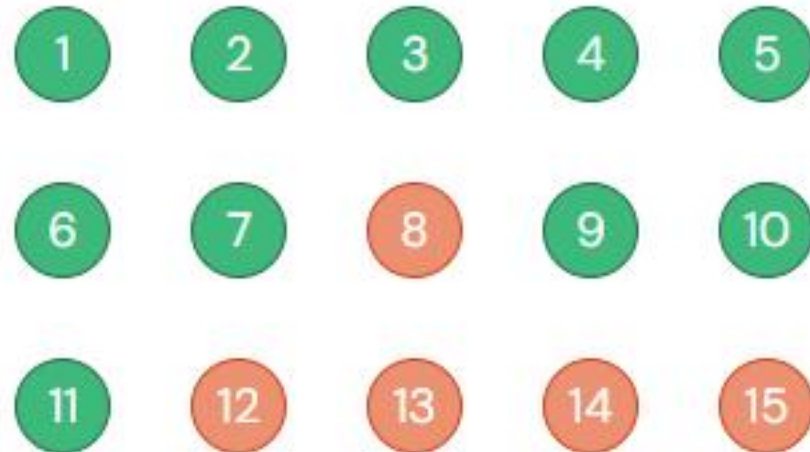
Time Left: 43:58

Section 1 of 3

AR



Finish Test



Answered	10
Revisit	5
Not Answered	5

Question # 13

✕ Revisit

Choose the option for which the analogy between the pair of words is the same as the one given in the question.
DELUGE : WATER ::

- ☐ A – sleet : hail
- ☐ B – dew : rain
- ☐ C – igloo : ice
- ☐ D – landslide : earth



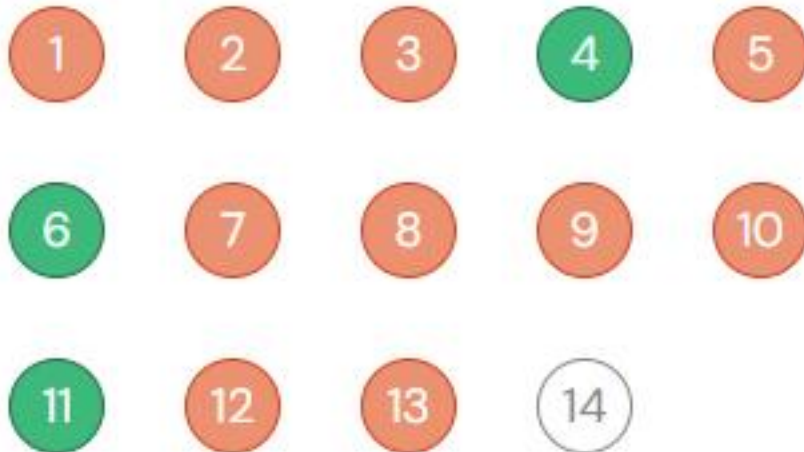
13/15



Time Left: 41:01

Section 2 of 3

OOPS



Answered	3
Revisit	10
Not Answered	11

Question # 1

Revisit

Which of the following is not a type of Inheritance:

A)Multiple

B)Multilevel

C)Hybrid

- ☐ A – Both A and B
- ☐ B – Both B and C
- ☐ C – Both A and C
- ☐ D – None of the above options



1/14



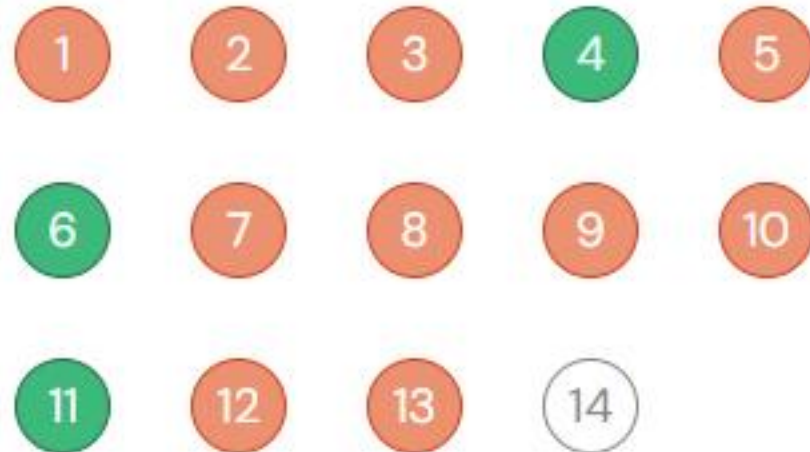
Time Left: 40:41

Section 2 of 3

OOPS



Finish Test



Answered	3
Revisit	10
Not Answered	11

Question # 3

✕ Revisit

In which of the following process the programmer tries to ensure that the entity is named in a manner that will make sense and that it will have all the relevant aspects included and none of the extraneous ones

- ☐ A – inheritance
- ☐ B – Encapsulation
- ☐ C – Abstraction
- ☐ D – Polymorphism



3/14



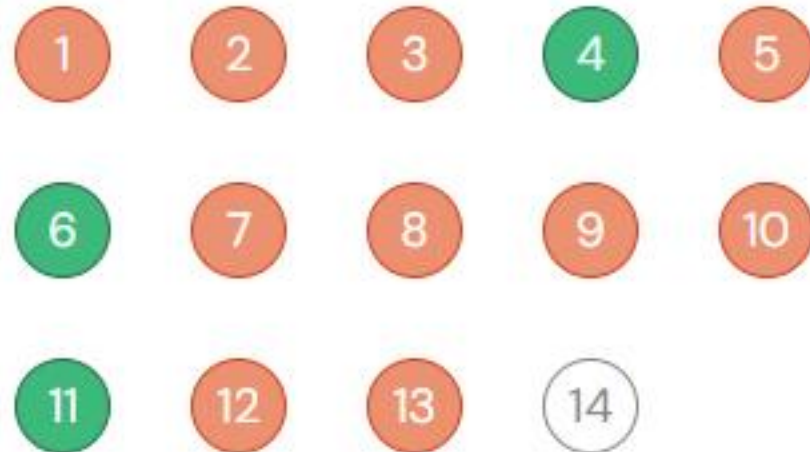
Time Left: 40:34

Section 2 of 3

OOPS



Finish Test



Answered	3
Revisit	10
Not Answered	11

Question # 5

✕ Revisit

Which of the following OOP concept is an attribute of an object, and it contains all data which is hidden. That hidden data can be restricted to the members of that class.

- ☐ A – Abstraction
- ☐ B – Encapsulation
- ☐ C – Polymorphism
- ☐ D – Inheritance



5/14



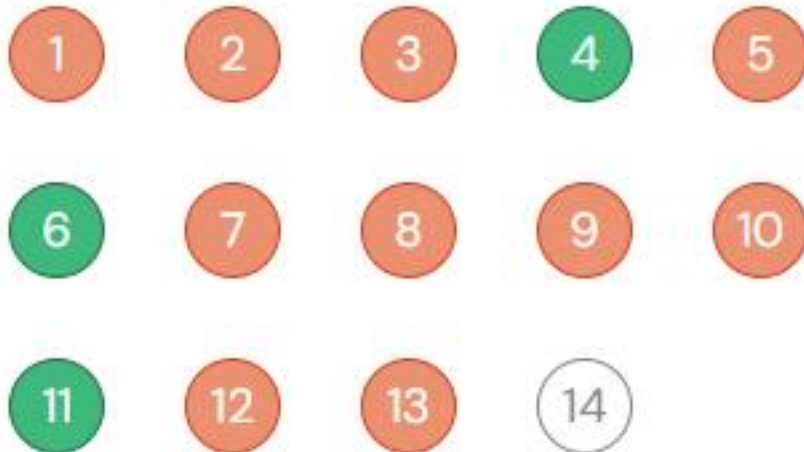
Time Left: 40:26

Section 2 of 3

OOPS



Finish Test



Answered	3
Revisit	10
Not Answered	11

Question # 7

✕ Revisit

Which of the following is a "using" relationship between two or more objects in which the objects have their own life time and there is no owner?

- ☐ A – Association
- ☐ B – Aggregation
- ☐ C – Composition
- ☐ D – none of the following options



7/14



Time Left: 40:23

Section 2 of 3

OOPS



Finish Test



Answered	3
Revisit	10
Not Answered	11

Question # 8

✕ Revisit

A user needs to send an email to somebody and the only information(s) he needs to know is the email id of the person to whom the email has to be sent. He does not need to know any of the details like how the code of sending email works, how it is sent across the network, how it is delivered to the intended person. To which OOPS concept is this example related to?

- ☐ A – Inheritance
- ☐ B – Abstraction
- ☐ C – Encapsulation
- ☐ D – Polymorphism



8/14



Time Left: 40:52

Section 2 of 3

OOPS



Finish Test



Answered	3
Revisit	10
Not Answered	11

Question # 2

✕ Revisit

Which of the following is the process of taking away or removing characteristics from something in order to reduce it to a set of essential characteristics

- ☐ A – inheritance
- ☐ B – Encapsulation
- ☐ C – Abstraction
- ☐ D – Polymorphism



2/14



Time Left: 39:55

Section 2 of 3

OOPS



Finish Test



Answered	3
Revisit	10
Not Answered	11

Question # 9

✕ Revisit

Which of the following is a "using" relationship between two or more objects in which the objects have their own life time and there is no owner?

- ☐ A – Association
- ☐ B – Aggregation
- ☐ C – Composition
- ☐ D – none of the following options



9/14



Time Left: 39:36

Section 2 of 3

OOPS



Finish Test



Answered	3
Revisit	10
Not Answered	11

Question # 10

✕ Revisit

Which of the following statements about a protected derivation of a derived class from a base class is true?

- ☐ A – Private members of the base class become protected members of the derived class.
- ☐ B – Protected members of the base class become public members of the derived class.
- ☐ C – Public members of the base class become protected members of the derived class.
- ☐ D – Protected derivation does not affect private and protected members of the derived class.



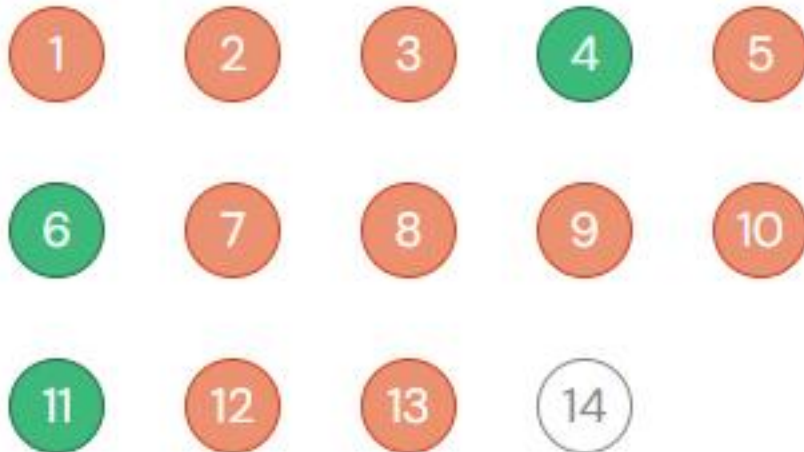
10/14



Time Left: 39:22

Section 2 of 3

OOPS



Answered	3
Revisit	10
Not Answered	11

Question # 12

Revisit

Which of the following can be declared as private?

- a) Constructor
- b) Static Methods
- c) Abstract methods

- ☐ A – Only a
- ☐ B – Both a and c
- ☐ C – Only b
- ☐ D – Only c



12/14



Time Left: 39:20

Section 2 of 3

OOPS



Finish Test



Answered	3
Revisit	10
Not Answered	11

Question # 13

✕ Revisit

Which of the following modifier is used to specify whether a member is a class member or an instance member?

- ☐ A – Public
- ☐ B – Protected
- ☐ C – Static
- ☐ D – Private



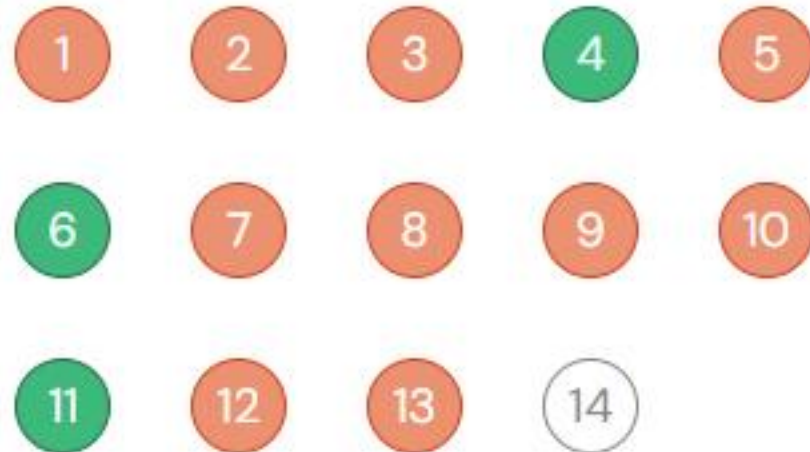
13/14



Time Left: 39:10

Section 2 of 3

OOPS



Answered	3
Revisit	10
Not Answered	11

Question # 14

Revisit

Which of the following access modifier can be accessed with in a class?

- a) Private
- b) Protected
- c) Public

- ☐ A – Both a and b
- ☐ B – Both a and c
- ☐ C – Both b and c
- ☐ D – None of the above options



14/14



Time Left: 39:58

Section 2 of 3

OOPS



Answered	3
Revisit	10
Not Answered	11

Finish Test

Question # 8

✕ Revisit

A user needs to send an email to somebody and the only information(s) he needs to know is the email id of the person to whom the email has to be sent. He does not need to know any of the details like how the code of sending email works, how it is sent across the network, how it is delivered to the intended person. To which OOPS concept is this example related to?

- ☐ A – Inheritance
- ☐ B – Abstraction
- ☐ C – Encapsulation
- ☐ D – Polymorphism



8/14



Time Left: 28:17

Section 2 of 3

OOPS



Finish Test



Answered	10
Revisit	4
Not Answered	4

Question # 9

Revisit

Which of the following is a "using" relationship between two or more objects in which the objects have their own life time and there is no owner?

- ☐ A – Association
- ☐ B – Aggregation
- ☐ C – Composition
- ☐ D – none of the following options



9/14



Time Left: 28:12

Section 2 of 3

OOPS



Finish Test



Answered	10
Revisit	4
Not Answered	4

Question # 13

✕ Revisit

Which of the following modifier is used to specify whether a member is a class member or an instance member?

- ☐ A – Public
- ☐ B – Protected
- ☐ C – Static
- ☐ D – Private



13/14



Time Left: 28:34

Section 2 of 3

OOPS



Answered	10
Revisit	4
Not Answered	4

Question # 1

Revisit

Which of the following is not a type of Inheritance:

A)Multiple

B)Multilevel

C)Hybrid

- ☐ A – Both A and B
- ☐ B – Both B and C
- ☐ C – Both A and C
- ☐ D – None of the above options



1/14



Time Left: 25:20

Section 3 of 3

Data Structure [Finish Test](#)

Question # 2

[✕ Revisit](#)

Consider an adjacency matrix representation of an undirected graph with n vertices and m edges. How long does it take, in the worst case, to check if there exists a path between two given vertices S and T ?

Note: Identify the tightest upper bound.

- ☐ A – $O(m * n)$
- ☐ B – $O(n^2)$
- ☐ C – $O(m + n)$
- ☐ D – $O(m + n \log n)$

Answered

4

Revisit

12

Not Answered

12



2/16



Time Left: 25:11

Finish Test

Section 3 of 3

Data Structure 

Answered	4
Revisit	12
Not Answered	12

Question # 3

 Revisit

Given a binary tree with n nodes and assuming $\text{size}(x)$ denotes the number of nodes in the sub tree rooted at the node x . How long does it take, in the worst case, to compute $\text{size}(x)$ for every node x of the tree?

Note: Choose the tightest upper bound.

- ☐ A – $O(n)$
- ☐ B – $O(n \log n)$
- ☐ C – $O(\text{height})$
- ☐ D – $O(n^2)$



3/16



Time Left: 25:00

Section 3 of 3

Data Structure [Finish Test](#)

Answered	4
Revisit	12
Not Answered	12

Question # 4

[✕ Revisit](#)

If $A[x][y]$ is a 2-D array, B is base address and k is size of 1 block, then what is the address of element at $A[i][j]$ if elements are arranged as column wise?

- ☐ A – $B + (i * k) + (j * x * k)$
- ☐ B – $B + (j * k) + (i * y * k)$
- ☐ C – $B + (j * k) + (i * x * k)$
- ☐ D – $B + (j * y * k) + (i * x * k)$



4/16



Time Left: 25:27

Finish Test

Section 3 of 3

Data Structure 

Answered	4
Revisit	12
Not Answered	12

Question # 1

 Revisit

Consider a uniform hash function to hash n distinct keys into an array T of length m , meaning each key is mapped independently and uniformly to a random bucket. What is the expected number of keys that get mapped to the first bucket?

- ☐ A – m/n
- ☐ B – $1/n$
- ☐ C – n/m
- ☐ D – $1/m$



1/16



Time Left: 24:52

Section 3 of 3

Data Structure [Finish Test](#)

Answered	4
Revisit	12
Not Answered	12

Question # 7

[✕ Revisit](#)What is the default value of array data type `wchar_t`?

- ☐ A – The data type `wchar_t` does not exist.
- ☐ B – 1
- ☐ C – 0
- ☐ D – Blank



7/16



Time Left: 24:48

Section 3 of 3

Data Structure [Finish Test](#)

Answered	4
Revisit	12
Not Answered	12

Question # 8

[✕ Revisit](#)

Analysis of time complexity remains an open problem for which of the following sorting algorithms?

- ☐ A – Quick sort
- ☐ B – Tim sort
- ☐ C – Shell sort
- ☐ D – None of the given options



8/16



Time Left: 24:42

Section 3 of 3

Data Structure [Finish Test](#)

Answered	4
Revisit	12
Not Answered	12

Question # 11

[✕ Revisit](#)

Suppose a program has the declaration as 'float sub_mark[5]', and the address of sub_mark[3] is 8091. What would be the address of sub_mark[1]?

- ☐ A – 8095
- ☐ B – 8083
- ☐ C – Float is not a correct datatype for array.
- ☐ D – 8087



11/16



Time Left: 24:38

Section 3 of 3

Data Structure [Finish Test](#)

Answered	4
Revisit	12
Not Answered	12

Question # 13

[✕ Revisit](#)

You have 3 sorted files f_1 , f_2 , f_3 having lengths 15, 10, 5 respectively. X is the minimum number of record moves required to merge these 3 files into single file. Then the value of x is:

- ☐ A – 55
- ☐ B – 40
- ☐ C – 45
- ☐ D – 30



13/16



Time Left: 24:33

Section 3 of 3

Data Structure [Finish Test](#)

Answered	4
Revisit	12
Not Answered	12

Question # 14

[✕ Revisit](#)

Which of the following options is/are a hash function?

- ☐ A – Folding method
- ☐ B – Division method
- ☐ C – All of the given options
- ☐ D – Mid square method



14/16



Time Left: 24:29

Section 3 of 3

Data Structure [Finish Test](#)

Answered	4
Revisit	12
Not Answered	12

Question # 15

[✕ Revisit](#)

Consider a 2-D array $A[5 \dots 10][5 \dots 10]$ stored in row-major order, where each element $A[i][j]$, $5 \leq i \leq 10$ and $5 \leq j \leq 10$ is of 4 bytes and the address of $A[5][5]$ is 1000. What is the address of $A[7][9]$?

- ☐ A – 1108
- ☐ B – 1068
- ☐ C – 1104
- ☐ D – 1064



15/16



Time Left: 24:24

Section 3 of 3

Data Structure [Finish Test](#)

Answered	4
Revisit	12
Not Answered	12

Question # 16

[✕ Revisit](#)

What would be the output of the code given? `#include<stdio.h>`

```
int main() {  
    char a;  
    char clist[10]= {1,2,3,4,5,6,7,8};  
    a=(clist+1)[5];  
    printf(" %d", a); return 0;  
}
```

- ☐ A – 7
- ☐ B – Program crashes
- ☐ C – Compilation error
- ☐ D – 6



16/16



Time Left: 24:54

Section 3 of 3

Data Structure [Finish Test](#)

Answered	4
Revisit	12
Not Answered	12

Question # 5

[✕ Revisit](#)

If the address of the first node of a linked list consisting of an integer and a pointer to next node is 1000, what will be the address of the second node of the same linked list?

- ☐ A – 1006
- ☐ B – None of the given options
- ☐ C – 1002
- ☐ D – 1008



5/16



Time Left: 23:32

Section 2 of 3

OOPS



Answered	10
Revisit	3
Not Answered	4

Question # 1

[✕ Revisit](#)

Which of the following is not a type of Inheritance:

A)Multiple

B)Multilevel

C)Hybrid

- ☐ A – Both A and B
- ☐ B – Both B and C
- ☐ C – Both A and C
- ☐ D – None of the above options



1/14



Section 2 of 3

OOPS



Answered	12
Revisit	2
Not Answered	2

Question # 1

Revisit

Which of the following is not a type of Inheritance:

A)Multiple

B)Multilevel

C)Hybrid

- ☐ A – Both A and B
- ☐ B – Both B and C
- ☐ C – Both A and C
- ☐ D – None of the above options



1/14



Time Left: 15:37

Section 3 of 3

Data Structure [Finish Test](#)

Answered	4
Revisit	12
Not Answered	12

Question # 2

[✕ Revisit](#)

Consider an adjacency matrix representation of an undirected graph with n vertices and m edges. How long does it take, in the worst case, to check if there exists a path between two given vertices S and T ?
Note: Identify the tightest upper bound.

- ☐ A – $O(m * n)$
- ☐ B – $O(n^2)$
- ☐ C – $O(m + n)$
- ☐ D – $O(m + n \log n)$



2/16



Time Left: 15:32

Section 3 of 3

Data Structure [Finish Test](#)

Answered	4
Revisit	12
Not Answered	12

Question # 3

[✕ Revisit](#)

Given a binary tree with n nodes and assuming $\text{size}(x)$ denotes the number of nodes in the sub tree rooted at the node x . How long does it take, in the worst case, to compute $\text{size}(x)$ for every node x of the tree?
Note: Choose the tightest upper bound.

- ☐ A – $O(n)$
- ☐ B – $O(n \log n)$
- ☐ C – $O(\text{height})$
- ☐ D – $O(n^2)$



3/16



Time Left: 15:26

Section 3 of 3

Data Structure [Finish Test](#)

Answered	4
Revisit	12
Not Answered	12

Question # 4

[✕ Revisit](#)

If $A[x][y]$ is a 2-D array, B is base address and k is size of 1 block, then what is the address of element at $A[i][j]$ if elements are arranged as column wise?

- ☐ A – $B + (i * k) + (j * x * k)$
- ☐ B – $B + (j * k) + (i * y * k)$
- ☐ C – $B + (j * k) + (i * x * k)$
- ☐ D – $B + (j * y * k) + (i * x * k)$



4/16



Time Left: 15:22

Section 3 of 3

Data Structure [Finish Test](#)

Answered	4
Revisit	12
Not Answered	12

Question # 5

[✕ Revisit](#)

If the address of the first node of a linked list consisting of an integer and a pointer to next node is 1000, what will be the address of the second node of the same linked list?

- ☐ A – 1006
- ☐ B – None of the given options
- ☐ C – 1002
- ☐ D – 1008



5/16



Time Left: 15:18

Section 3 of 3

Data Structure [Finish Test](#)

Answered	4
Revisit	12
Not Answered	12

Question # 7

[✕ Revisit](#)What is the default value of array data type `wchar_t`?

- ☐ A – The data type `wchar_t` does not exist.
- ☐ B – 1
- ☐ C – 0
- ☐ D – Blank



7/16



Time Left: 15:15

Section 3 of 3

Data Structure [Finish Test](#)

Answered	4
Revisit	12
Not Answered	12

Question # 8

[✕ Revisit](#)

Analysis of time complexity remains an open problem for which of the following sorting algorithms?

- ☐ A – Quick sort
- ☐ B – Tim sort
- ☐ C – Shell sort
- ☐ D – None of the given options



8/16



Time Left: 15:12

Section 3 of 3

Data Structure [Finish Test](#)

Answered	4
Revisit	12
Not Answered	12

Question # 11

[✕ Revisit](#)

Suppose a program has the declaration as 'float sub_mark[5]', and the address of sub_mark[3] is 8091. What would be the address of sub_mark[1]?

- ☐ A – 8095
- ☐ B – 8083
- ☐ C – Float is not a correct datatype for array.
- ☐ D – 8087



11/16



Time Left: 15:08

Section 3 of 3

Data Structure [Finish Test](#)

Answered	4
Revisit	12
Not Answered	12

Question # 13

[✕ Revisit](#)

You have 3 sorted files f1, f2, f3 having lengths 15, 10, 5 respectively. X is the minimum number of record moves required to merge these 3 files into single file. Then the value of x is:

- ☐ A – 55
- ☐ B – 40
- ☐ C – 45
- ☐ D – 30



13/16



Time Left: 15:05

Section 3 of 3

Data Structure [Finish Test](#)

Answered	4
Revisit	12
Not Answered	12

Question # 14

[✕ Revisit](#)

Which of the following options is/are a hash function?

- ☐ A – Folding method
- ☐ B – Division method
- ☐ C – All of the given options
- ☐ D – Mid square method



14/16



Time Left: 15:02

Section 3 of 3

Data Structure [Finish Test](#)

Answered	4
Revisit	12
Not Answered	12

Question # 15

[✕ Revisit](#)

Consider a 2-D array $A[5...10][5...10]$ stored in row-major order, where each element $A[i][j]$, $5 \leq i \leq 10$ and $5 \leq j \leq 10$ is of 4 bytes and the address of $A[5][5]$ is 1000. What is the address of $A[7][9]$?

- ☐ A – 1108
- ☐ B – 1068
- ☐ C – 1104
- ☐ D – 1064



15/16



Time Left: 15:41

Section 3 of 3

Data Structure [Finish Test](#)

Answered	4
Revisit	12
Not Answered	12

Question # 1

[✕ Revisit](#)

Consider a uniform hash function to hash n distinct keys into an array T of length m , meaning each key is mapped independently and uniformly to a random bucket. What is the expected number of keys that get mapped to the first bucket?

- ☐ A – m/n
- ☐ B – $1/n$
- ☐ C – n/m
- ☐ D – $1/m$



1/16



Time Left: 14:58

Section 3 of 3

Data Structure [Finish Test](#)

Answered	4
Revisit	12
Not Answered	12

Question # 16

[✕ Revisit](#)

What would be the output of the code given? `#include<stdio.h>`

```
int main() {  
    char a;  
    char clist[10]= {1,2,3,4,5,6,7,8};  
    a=(clist+1)[5];  
    printf(" %d", a); return 0;  
}
```

- ☐ A – 7
- ☐ B – Program crashes
- ☐ C – Compilation error
- ☐ D – 6



16/16



Time Left: 07:07

Section 2 of 3

OOPS



Answered	12
Revisit	2
Not Answered	2

Question # 1

[✕ Revisit](#)

Which of the following is not a type of Inheritance:

- A)Multiple
- B)Multilevel
- C)Hybrid

- ☐ A – Both A and B
- ☐ B – Both B and C
- ☐ C – Both A and C
- ☐ D – None of the above options

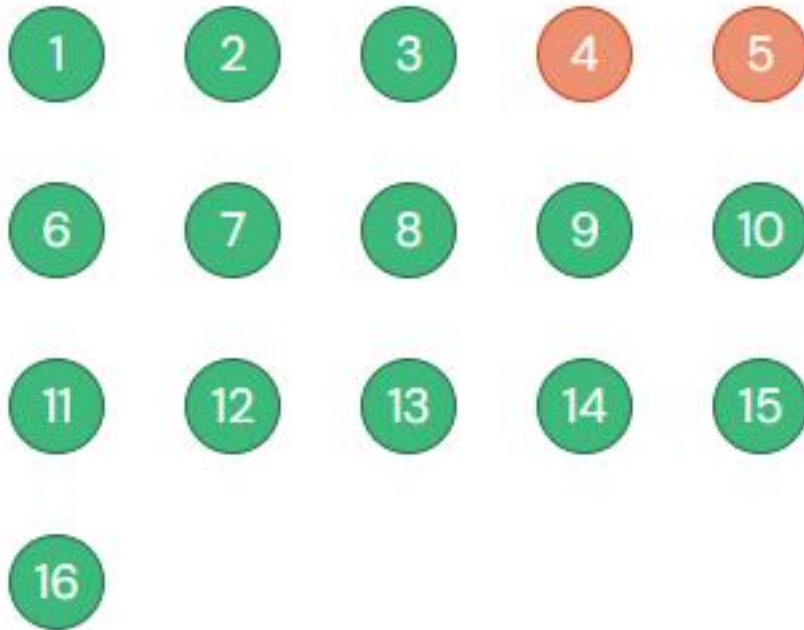


1/14



Time Left: 06:47

Section 3 of 3

Data Structure [Finish Test](#)

Answered	14
Revisit	2
Not Answered	2

Question # 4

[✕ Revisit](#)

If $A[x][y]$ is a 2-D array, B is base address and k is size of 1 block, then what is the address of element at $A[i][j]$ if elements are arranged as column wise?

- ☐ A – $B + (i * k) + (j * x * k)$
- ☐ B – $B + (j * k) + (i * y * k)$
- ☐ C – $B + (j * k) + (i * x * k)$
- ☐ D – $B + (j * y * k) + (i * x * k)$

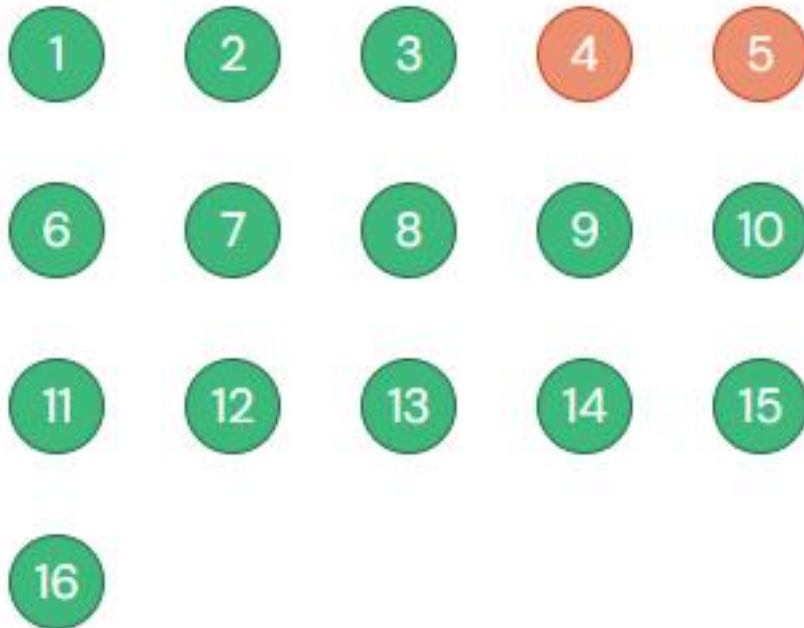


4/16



Time Left: 06:53

Section 3 of 3

Data Structure [Finish Test](#)

Answered	14
Revisit	2
Not Answered	2

Question # 5

[✕ Revisit](#)

If the address of the first node of a linked list consisting of an integer and a pointer to next node is 1000, what will be the address of the second node of the same linked list?

- ☐ A – 1006
- ☐ B – None of the given options
- ☐ C – 1002
- ☐ D – 1008



5/16





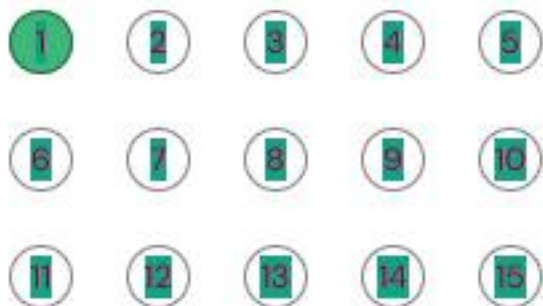
Time Left: 58:07

Finish Test

Revisit

Section 1 of 3

AR



Answered

1

Revisit

0

Not Answered

14

Question # 2

In a certain coding system 'BACK' is coded as '21311'. How will you code 'ENGAGE'?

- ☐ A - 5147175
- ☐ B - 5475171
- ☐ C - 4136175
- ☐ D - 547175
- ☐ E - 574157



2/15



Time Left: 57:00

Finish Test

Revisit

Section 1 of 3

AR



Answered	2
Revisit	0
Not Answered	13

Question # 3

The problem below consists of a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statements is sufficient to answer the question. Read both the statements and give the answer:

How many children are there in the group?

I. Sangita has scored more marks than 12 children in the group.

II. Reena has scored less than Sangita.

- ☐ A - Statement i alone is sufficient
- ☐ B - Statements i and ii together are sufficient.
- ☐ C - Either of the statements i or ii alone is sufficient.
- ☐ D - Statement ii alone is sufficient
- ☐ E - Statements i and ii together also are not sufficient.



3/15



Time Left: 55:43

Section 1 of 3

AR

Finish Test



Answered	5
Revisit	0
Not Answered	10

Question # 6

[Revisit](#)

Fill in the blank with the best suited word.

The game of chess is an example of a information system: the pieces sit inertly on the board until the players move them according to known rules.

- ☐ A - Synergistic
- ☐ B - Innovative
- ☐ C - Logistic
- ☐ D - Passive



6/15



Time Left: 54:52

Finish Test

Section 1 of 3

AR



Answered	6
Revisit	0
Not Answered	9

Question # 7

Revisit

Fill in the blank with the best suited word.

A businessman must _____(i)_____ his horizons; a/an _____(ii)_____ attitude will get you nowhere in this age of global communications.

- ☐ A - (i) Broaden ; (ii) Bigoted
- ☐ B - (i) Restrain ; (ii) Extensive
- ☐ C - (i) Envisage ; (ii) Middling



7/15



Time Left: 53:54

Finish Test

Section 1 of 3

AR



Answered	8
Revisit	0
Not Answered	7

Question # 9

Revisit

A dog runs 4 miles toward north and turns right, runs 1 mile again, turns to its right, runs another 5 miles. It then, turns left and runs 2 miles; turns to its left and runs 5 miles, and finally turns to its right. What is the smallest distance between the dog's final point and the initial point, and in which direction is the dog facing?

- ☐ A - 5 miles, north-east
- ☐ B - 116 miles, north-east
- ☐ C - 116 miles, east
- ☐ D - 5 miles, east



9/15



Time Left: 54:12

Finish Test

Revisit

Question # 8

Choose the option for which the analogy between the pair of words is the same as the one given in the question.
INKBLOT : EYECART ::

- ☒ A - blotter : spectacles
- ☐ B - oculist : ophthalmologist
- ☐ C - psychiatrist : optometrist
- ☐ D - physician : specialist

Clear Response



8/15



Section 1 of 3

AR



Answered	7
Revisit	0
Not Answered	8

Time Left: 51:21

Section 1 of 3

AR



Finish Test



Answered	9
Revisit	0
Not Answered	6

Question # 10

Revisit

Choose the alternative that best completes the given comparison. Cataract: Eye :: ? : Lungs

- ☐ A - Trachoma
- ☐ B - Strabismus
- ☐ C - Tuberculosis
- ☐ D - Amblyopia



10/15



Time Left: 49:24

Finish Test

Section 1 of 3

AR



Answered	10
Revisit	0
Not Answered	5

Question # 11

 Revisit

Sam walks straight from point A to a point B, which is 6 km away from point A towards the east. He turns left, at 90 degrees and walks 10 km to C, where he turns left again at 90 degrees and walks 8 km to D. At D he turns left at 90 degrees and walks for 10 km to E. How far is (in km) A from E?

- ☐ A - 2 kms
- ☐ B - 1 km
- ☐ C - 8 kms
- ☐ D - 6 kms



11/15



Time Left: 47:26

Finish Test

Section 1 of 3

AR



Answered	11
Revisit	0
Not Answered	4

Question # 12

Revisit

The n th element of a series is represented as: $X_n = (-1)^n X_{n-1}$. If $X_0 = x$ and $x > 0$. Which of the following is always true?

- ☐ A - X_n is negative if n is odd.
- ☐ B - $X_{101} + X_{102} + X_{103} + \dots + X_{199} = x$
- ☐ C - X_n is positive if n is even
- ☐ D - $X_{1999} + X_{2000} = 2x$



12/15



Time Left: 45:58

Finish Test

Section 1 of 3

AR

- 1
- 2
- 3
- 4
- 5
- 6
- 7
- 8
- 9
- 10
- 11
- 12
- 13
- 14
- 15

Answered	12
Revisit	0
Not Answered	3

Question # 13

Revisit

Radha from point A walked 30Km toward north. Then moved 60km in south from here to reach point B. From here, she turned to her left and walked 10km to reach point C. From the point C, she walked 70km north and turned right to walk another 80km. Finally, she turned to her right and walked 70km before stopping to reach point D. What is the minimum distance between A and D?

- ☐ A - 30Km
- ☐ B - $30\sqrt{10}$ km
- ☐ C - $30\sqrt{15}$ km
- ☐ D - $30\sqrt{5}$ km



13/15



Time Left: 44:56

Section 1 of 3

AR



Finish Test



Answered	13
Revisit	0
Not Answered	2

Question # 14

Revisit

A person starts from point P in the east direction and moves 12 miles to the point Q. Then he turns right and moves 8 miles to the point R. Again, he turns right and moves 6 miles to the point S. He then moves 6 miles in the north to the point T. Finally, from there, he goes to his left for 6 miles to the point U. Which of the three points would form a triangle in which all the angles are less than 90 degree?

- ☐ A - TSR
- ☐ B - QTR
- ☐ C - UTS
- ☐ D - PTQ



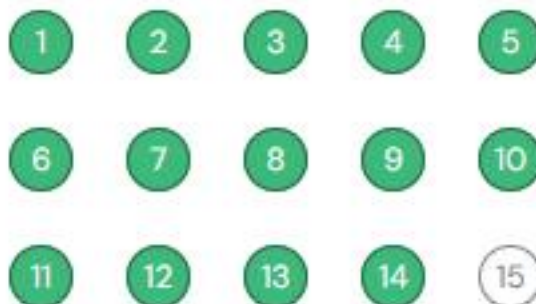
14/15



Time Left: 41:14

Section 1 of 3

AR



Answered	14
Revisit	0
Not Answered	1

Drone : Bee :: ? : Doe

- ☐ A - Stag
- ☐ B - Bird
- ☐ C - Tiger
- ☐ D - Hen



15/15



Time Left: 37:53

Finish Test

Revisit

Question # 4

Which of the following is a technique of making the fields in a class private and providing access to these fields with the help of public methods. If a field is declared private, it cannot be accessed by anyone outside the class, thereby **hiding the fields within the class**

- ☐ A - Abstraction
- ☐ B - Encapsulation
- ☐ C - Polymorphism
- ☐ D - Inheritance

Section 2 of 3

OOPS



Answered	3
Revisit	0
Not Answered	11



4/14



Time Left: 36:02

Finish Test

Revisit

Question # 9

Which of the following is a specialized form of association between two or more objects in which the objects have their own life-cycle but there exists an ownership as well?

- ☐ A - Abstraction
- ☐ B - Aggregation
- ☐ C - Composition
- ☐ D - none of the following options

Section 2 of 3

OOPS



Answered	8
Revisit	0
Not Answered	6



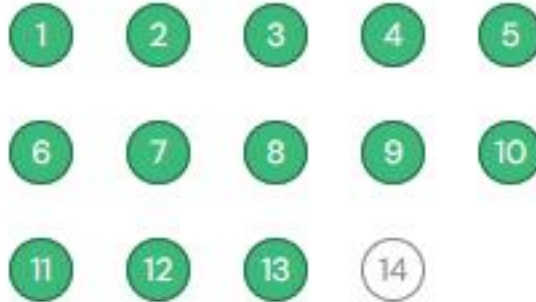
9/14



Time Left: 34:48

Section 2 of 3

OOPS



Answered	13
Revisit	0
Not Answered	1

Question # 14

Revisit

Which of the following can be declared as private?

- a) Constructor
- b) Static Methods
- c) Abstract methods

- ☐ A - Only a
- ☐ B - Both a and c
- ☐ C - Only b
- ☐ D - Only c



14/14



Time Left: 29:42

Finish Test

Section 2 of 3 Data Structure



Answered	2
Revisit	0
Not Answered	14

Question # 3

 Revisit

Given a binary tree with n nodes and assuming $\text{size}(x)$ denotes the number of nodes in the sub tree rooted at the node x . How long does it take, in the worst case, to compute $\text{size}(x)$ for every node x of the tree?

Note: Choose the tightest upper bound.

- ☐ A - $O(n)$
- ☐ B - $O(n \log n)$
- ☐ C - $O(\text{height})$
- ☐ D - $O(n^2)$



3/16



Time Left: 26:37

Finish Test

Revisit

Question # 6

A min-priority queue implemented over a binary heap can support deletion of element having smallest key at which complexity?

- ☐ A - $O(\log \log N)$
- ☒ B - $O(\log N)$
- ☐ C - $O(N)$
- ☐ D - $O(1)$

Clear Response

Section 2 of 3

Data Structure



Answered	5
Revisit	0
Not Answered	11



6/16



Time Left: 26:23

Finish Test

Section 2 of 3 Data Structure

- 1 2 3 4 5
- 6 7 8 9 10
- 11 12 13 14 15
- 16

Answered	7
Revisit	0
Not Answered	9

Question # 8

Revisit

Which of the following statements about a binary heap is false?

- A - The main reason that we can use an array to represent a binary heap is because the tree is full.
- B - In the worst case, deleting a maximum key from a max-heap containing N keys takes $\sim 2 \lg N$ comparisons.
- C - The main reason to implement a max-heap using an array is to improve the running time of insert and delete_max operations.
- D - Let $a[]$ be a binary heap that uses 0-based indexing. Then, the children of $a[k]$ are $a[2k+1]$ and $a[2k+2]$.



8/16



Time Left: 27:05

Finish Test

Section 2 of 3 Data Structure

- 1 2 3 4 5
- 6 7 8 9 10
- 11 12 13 14 15
- 16

Answered	4
Revisit	0
Not Answered	12

Question # 5

Revisit

Insertion of a node in a circular linked list requires modification of at the least:

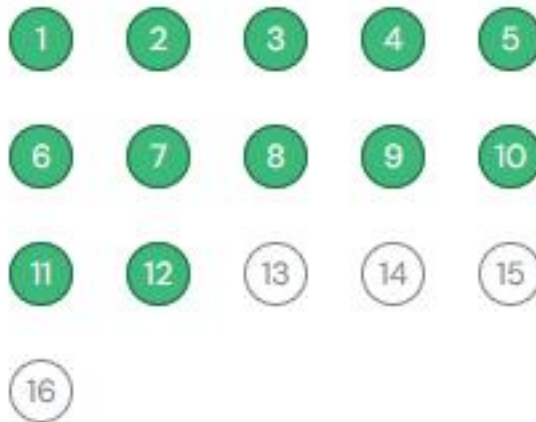
- ☐ A - 1 pointer
- ☐ B - 3 pointers
- ☐ C - No pointer
- ☒ D - 2 pointers

Clear Response

Time Left: 22:23

Finish Test

Section 2 of 3 Data Structure



Answered	12
Revisit	0
Not Answered	4

Question # 13

[Revisit](#)

An element is added in max heap of height h . The maximum time required to add an element into existing heap of n elements is:

- ☐ A - $O(h \log n)$
- ☐ B - $O(h)$
- ☐ C - $O(n)$
- ☐ D - $O(n \log n)$



13/16

